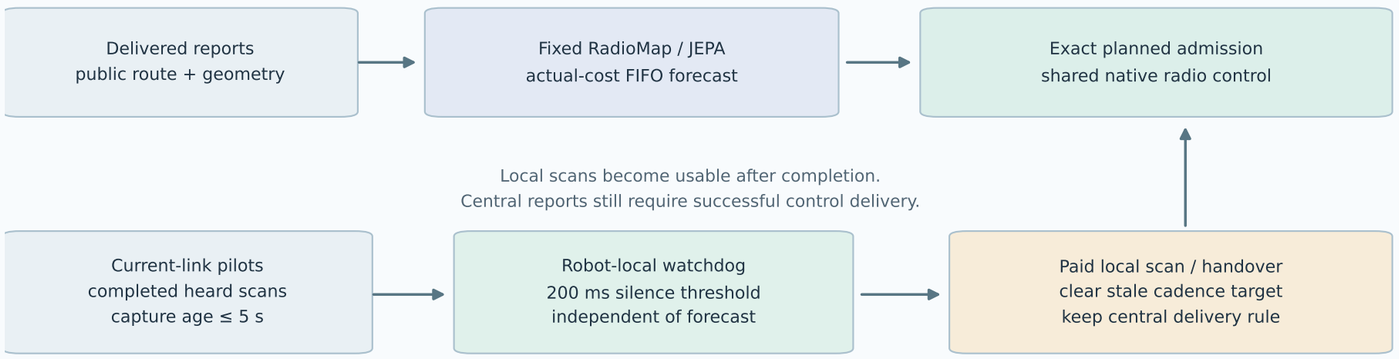
Local recovery, measured completion and the cost of acting

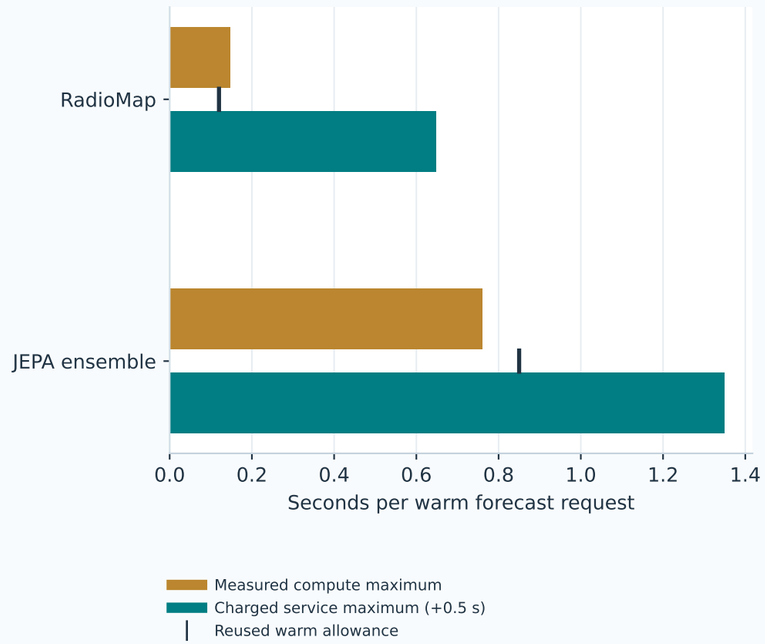
A causal control-path schematic beside measured confirmation traces; no new model fitting or physical-car qualification.

4 s development stopped: Map +5.505 pp; JEPA +4.748 pp. Its confirmation cases were not run.

A Local recovery does not wait for cloud delivery



B Warm forecast work and service charge

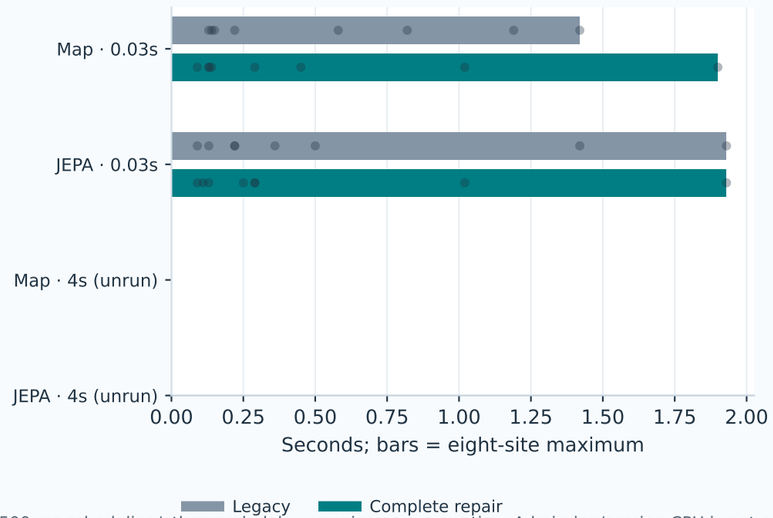


C Actual watchdog recovery and sensing

Family / cost	Watchdog moves	Due scans while silent	All moves legacy → repair
Map · 0.03 s	4.5	0.8	134.4 → 186.5
JEPA · 0.03 s	2.6	0.8	145.6 → 188.5
Map · 4 s	unrun	unrun	unrun
JEPA · 4 s	unrun	unrun	unrun

Counts are means per 100-robot, 60 s trial.
Only already-due scans can bypass delivery waiting;
90 ms scan outages and full handovers are paid.

D Longest observed control silence



E Paid-cost and replication evidence

Factual confirmation runs: 48
Actual forecast jobs: 2,880
Paid-completion failures: 0
Profile-exceeded cases: 1 (diagnostic)
Logged local watchdog checks: 167
Largest consumed forecast age: 17.98 s
Exact offline trace replays: 6 / 6
Fresh live-cost repetitions: 6 (no extra sites)
Fresh repeats with changed availability: 0
Fresh maximum absolute miss change: 0

Timing uses actual history-to-forecast issue work; startup/loading is separate. The shared 500 ms scheduling/other-work delay remains an assumption. Admission/scoring CPU is not separately scheduled.
Offline replays use original paid schedules and prove deterministic trace replay, not freshly measured deadlines. Fresh repetitions measure timing sensitivity without extra independent sites.
Native control consumes current pilots and completed, heard, capture-age-limited local scans. Model, sensing and handover effects form a controller package. Shared procedural CAD, conditional 60 s warm history and finite trials do not establish hardware safety or indefinite reliability.